



Lesson 3 — Regression

Technologies for Artificial Intelligence

Fabio Antonini — Università degli Studi
dell'Aquila

Before we start

- Exercise 2 was due today
- The pipeline you built is the one we fit models into from now on

Today: the first real model

- The model and what it claims
- The exact solution — and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

One dataset, all lesson

600 houses. Six measurements. A price.

And we know the coefficients that generated it.

What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up.

- Every square metre: the same 2,400 €
- Every kilometre out: the same –6,500 €

Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors fails: $+50,000$ and $-50,000$ cancel to zero.

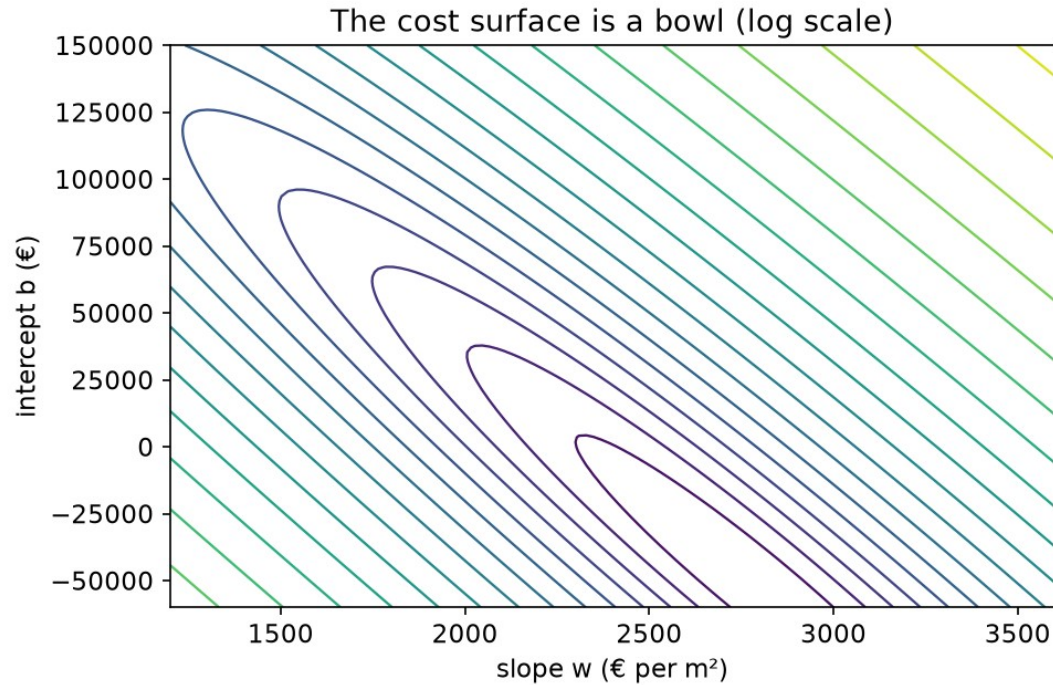
Why squared, not absolute

- **Differentiable everywhere** — no corner at zero
- **Punishes one large error more** than several small ones
- Under Gaussian noise, it **is** maximum likelihood

The cost function

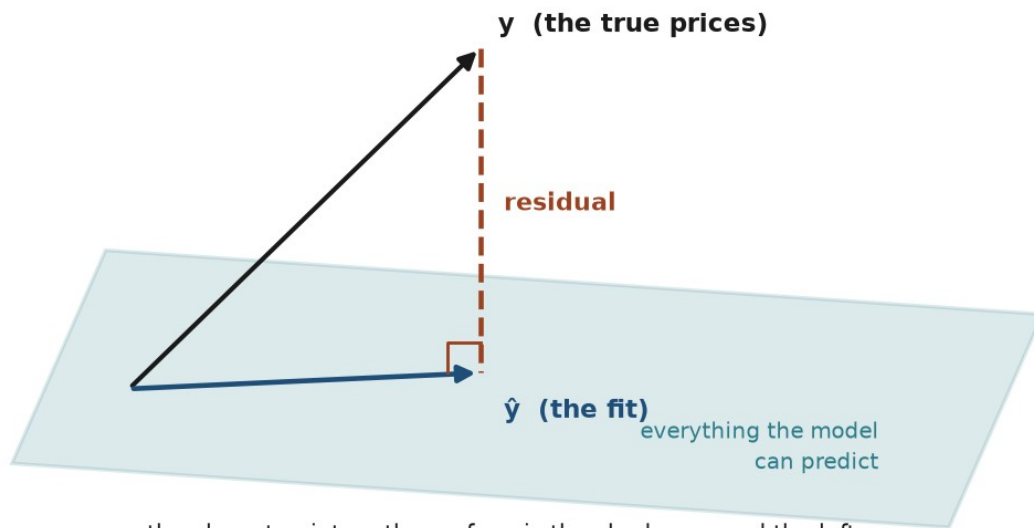
$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

The cost is a bowl



The picture behind the exact solution

Least squares is a projection



the closest point on the surface is the shadow — and the leftover error is perpendicular, or you could have done better

The normal equation

$$\theta = (X^T X)^{-1} X^T y$$

So why learn anything else?

- $(X^T X)^{-1}$ costs about n^3 operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

The picture behind gradient descent

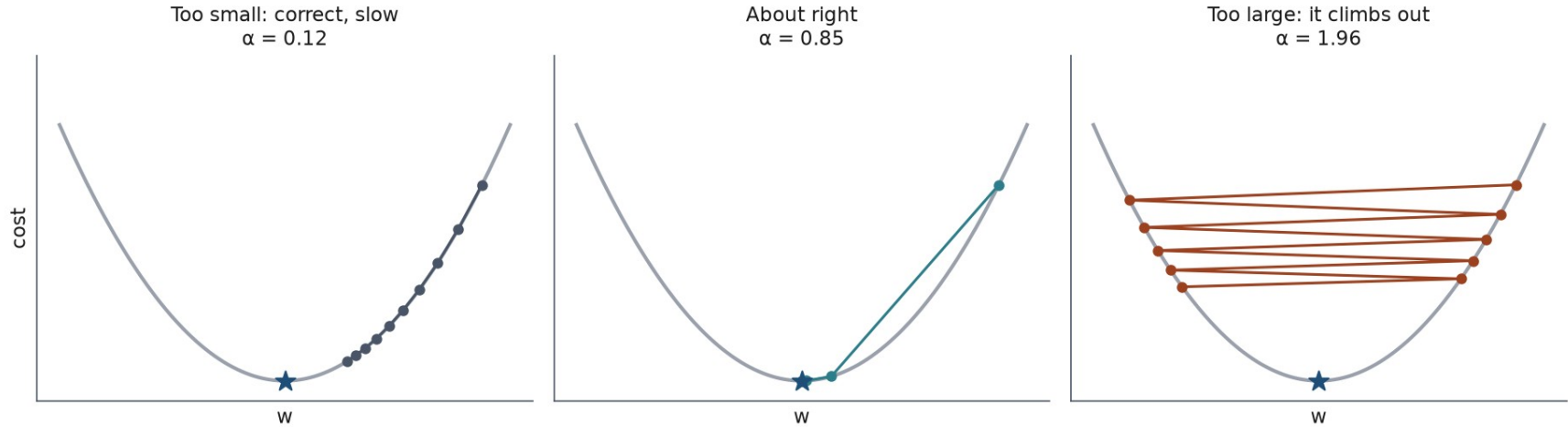
You are on a hillside in fog.

You cannot see the valley — but you can feel which way the ground slopes.

The update rule

$$w \leftarrow w - \alpha \frac{\partial J}{\partial w}, \quad \frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

Three learning rates



Notebook 1, live

The model, the cost, the exact solution, the iterative one — from scratch.

Break

What if the relationship bends?

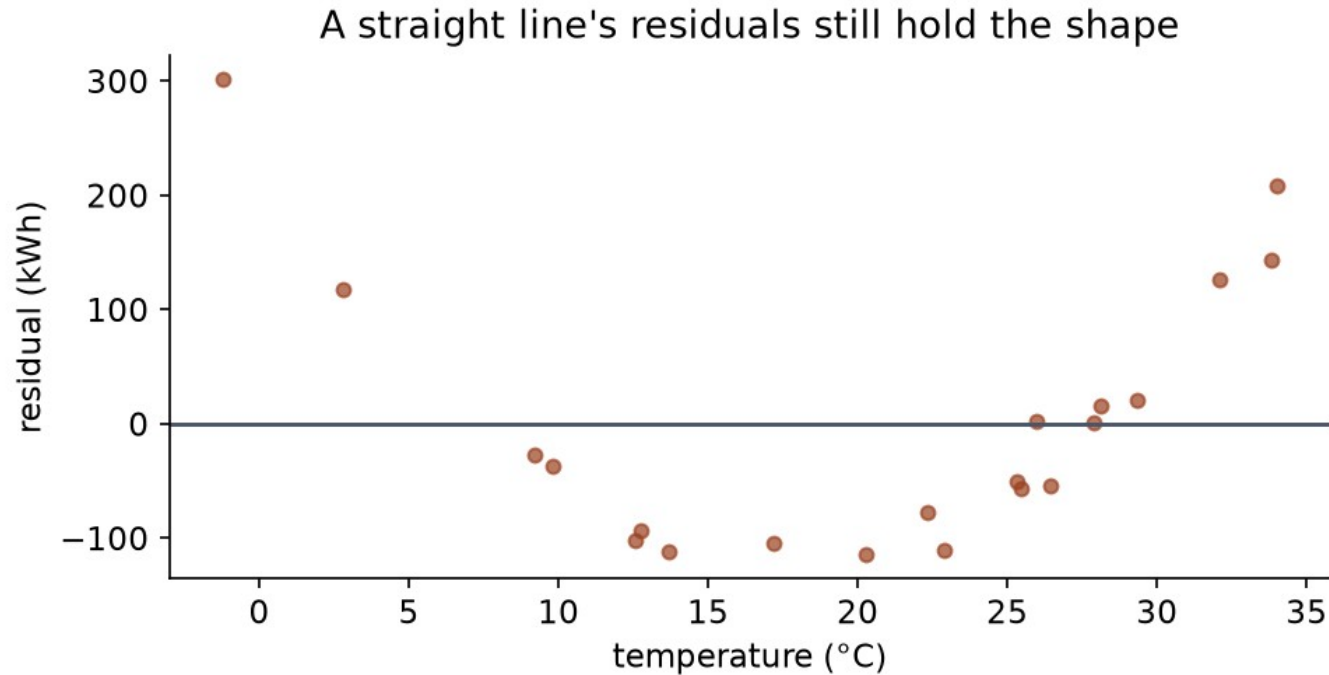
Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

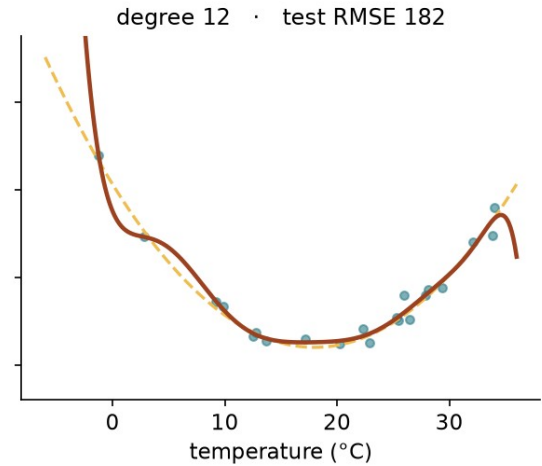
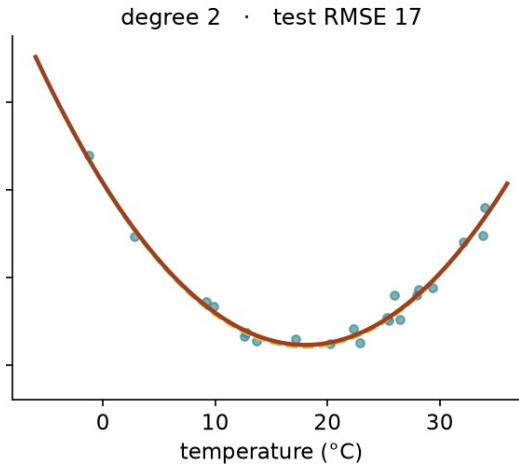
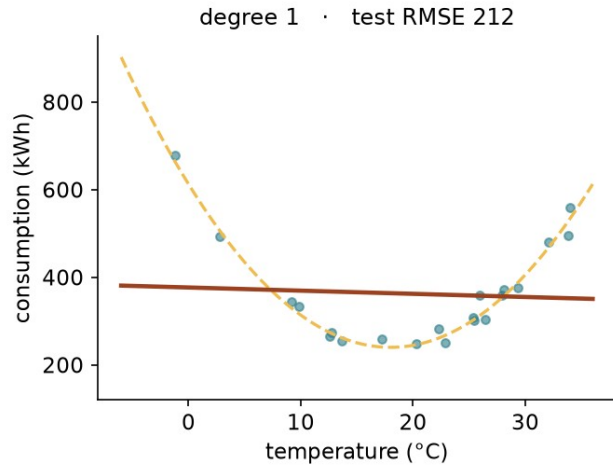
“Linear” means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

The straight line's residuals keep the shape



So use more flexibility?



The table that matters

Degree	Train RMSE	Test RMSE
1	113.3	212.1
3	17.5	16.9
9	16.5	118.2
12	16.3	182.0

Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **247,514**

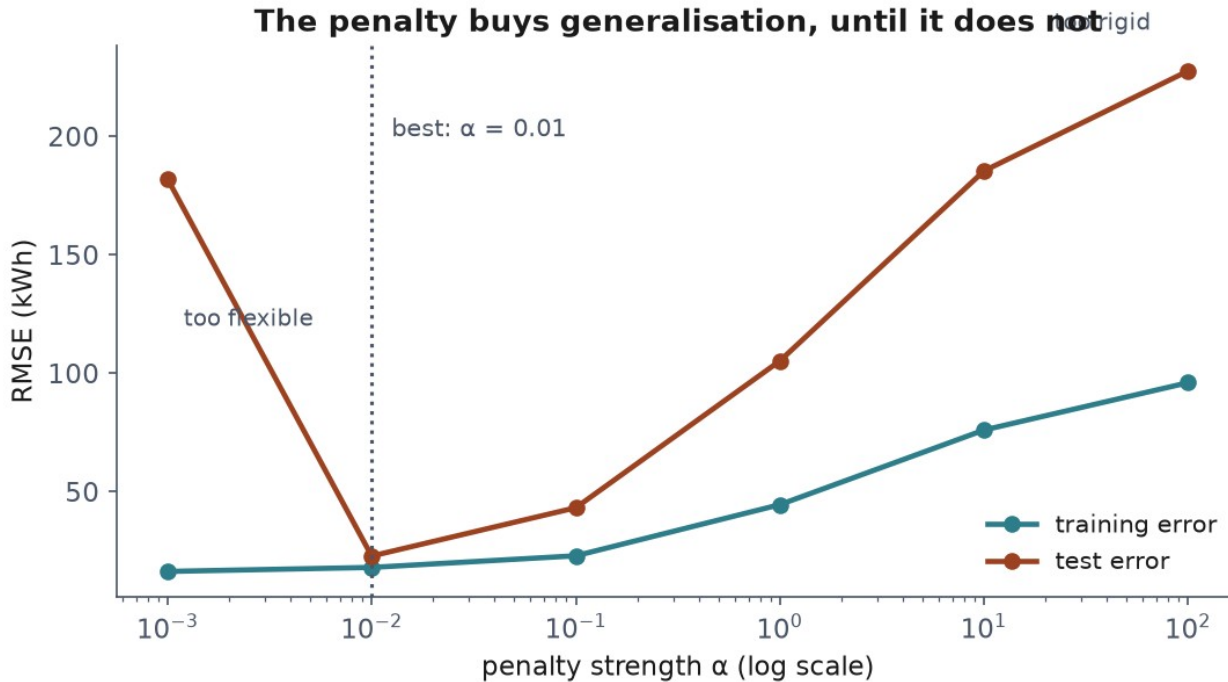
Charge for size

$$J_{\text{ridge}} = \text{MSE} + \alpha \sum_j w_j^2 \qquad J_{\text{lasso}} = \text{MSE} + \alpha \sum_j |w_j|$$

What a penalty buys

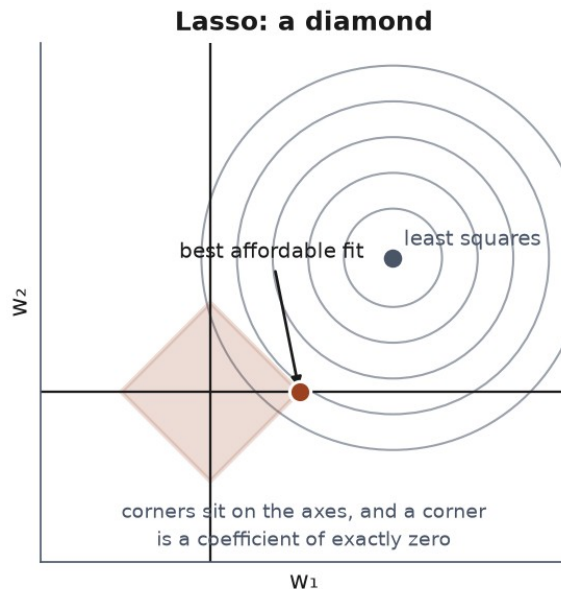
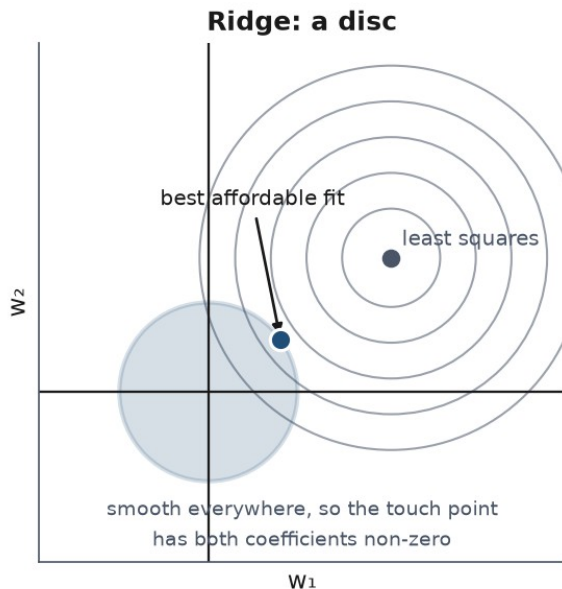
Model	Train	Test	max w
none	16.3	182.0	247,514
$\alpha = 0.01$	18.0	22.8	365
$\alpha = 1$	44.5	105.2	156
$\alpha = 100$	96.0	227.6	6

Too flexible, too rigid

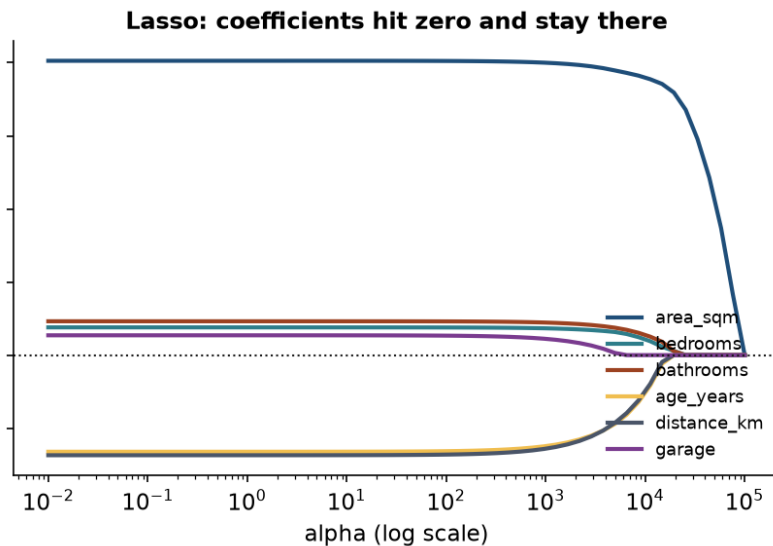
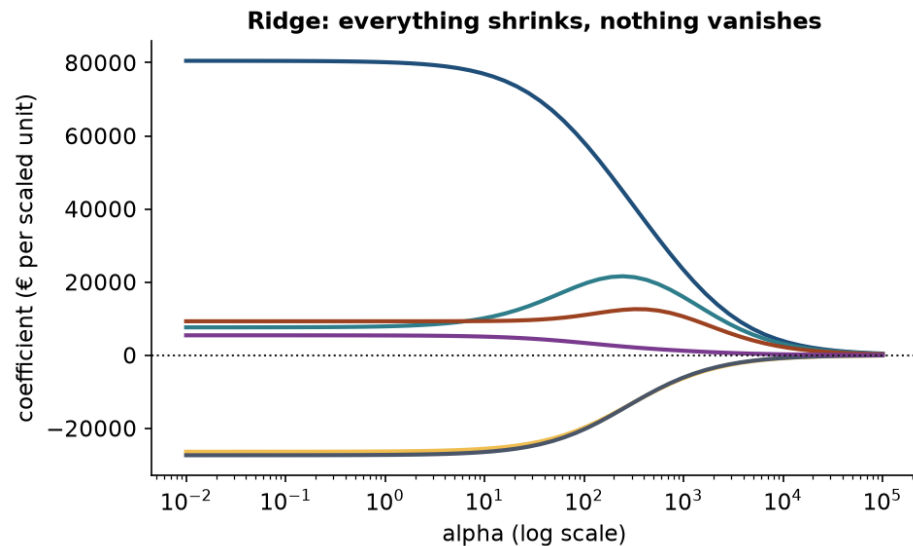


The picture behind Ridge vs Lasso

The budget you can afford has a shape



Ridge shrinks, Lasso selects



The case Ridge was invented for

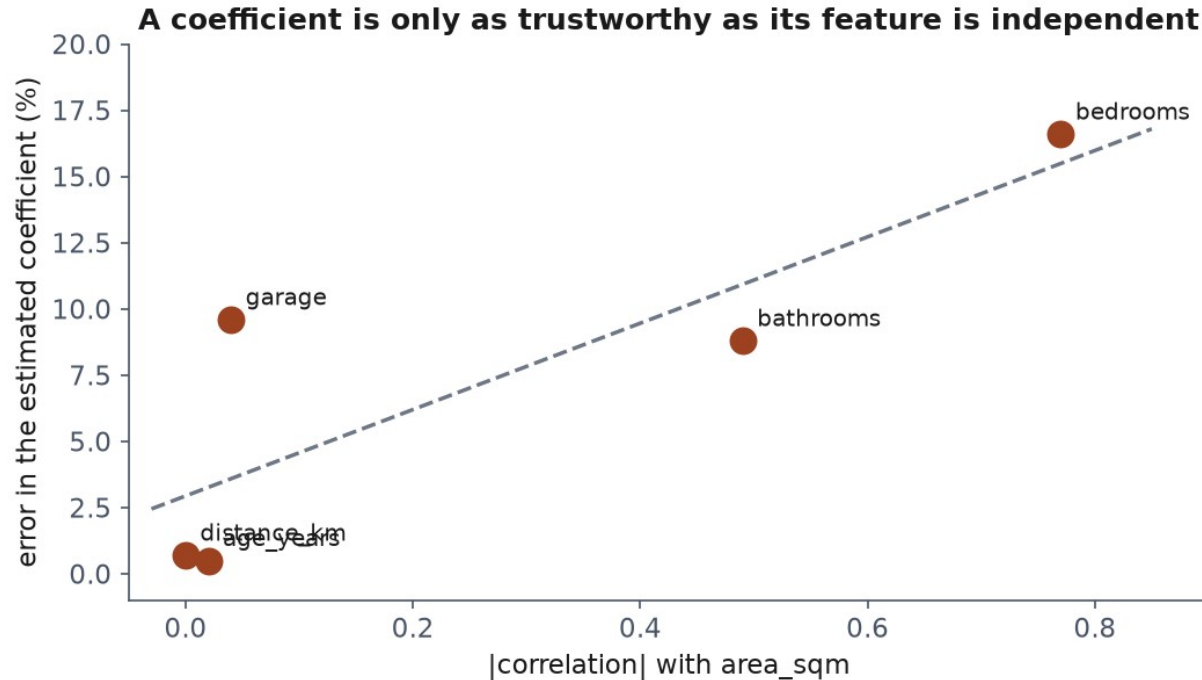
`area_sqm` and `area_sqft` — the same measurement, two units.

Correlation: **0.999999**

The coefficients are noise

Split	area_sqm	area_sqft
0	9,261	-640
1	14,535	-1,127
2	45,307	-3,989
3	39,061	-3,402

When can you trust a coefficient?



Notebook 3, live

Ridge on the disaster, the regularisation paths, the collinear case.

What we did today

- A model that claims a fixed amount per unit
- An exact solution — a projection — and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients

Homework — due Friday 16 October

Exercises/03_regression.md

Fit, regularise, and **explain which coefficients you believe.**